

IoT-Based Air Pollution Monitoring System for Real-Time Systems

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Abstract

The rapid increase in air pollution, particularly in urban environments, has created a critical need for efficient and real-time air quality monitoring systems. Traditional monitoring methods are often expensive, location-restricted, and lack real-time accessibility. This paper presents an Internet of Things (IoT)-based framework for real-time monitoring of environmental parameters including temperature, humidity, and harmful gases such as ammonia, methane, carbon monoxide, and ethanol. The system employs DHT11, MQ5, and MQ135 sensors interfaced with an Arduino microcontroller. Collected data is transmitted via Wi-Fi to the ThingSpeak cloud platform, enabling remote visualization and analysis. An alert mechanism using buzzer and LED indicators notifies users when pollutant concentrations exceed predefined safety thresholds. Experimental results demonstrate reliable data acquisition, efficient communication, and effective real-time monitoring capabilities. The proposed system offers a low-cost, scalable, and user-friendly solution for environmental monitoring that can contribute to improved public health awareness and support the development of smart, sustainable environments.

Keywords—Air quality monitoring, Arduino, Internet of Things, real-time monitoring, sensor networks, smart environment.

Introduction

Air pollution has emerged as one of the most critical environmental and public health challenges worldwide, particularly in rapidly urbanizing regions. Continuous exposure to polluted air containing harmful gases such as carbon monoxide (CO), methane (CH₄), ammonia (NH₃), and particulate matter leads to severe health issues including respiratory diseases, asthma, bronchitis, and long-term cardiovascular complications [1]. According to the World Health Organization, approximately 93% of children breathe polluted air

containing particulate matter of less than 10 micrometers, causing chronic diseases [2]. Hence, continuous monitoring of air quality has become essential to ensure safe and healthy living environments.

The rapid advancement of the Internet of Things has enabled the development of interconnected sensing devices capable of collecting real-time environmental data [3]. IoT systems consist of smart sensors, microcontrollers, and communication modules that work together to monitor and transmit data over the internet without human intervention. This facilitates remote monitoring, automation, and efficient data management. When combined with cloud platforms, IoT systems support real-time analysis, visualization, and decision-making capabilities [4].

Traditional air quality monitoring systems are often expensive, bulky, and restricted to specific locations, limiting their accessibility and widespread deployment [5]. Moreover, most conventional systems primarily focus on data collection and lack real-time alert mechanisms and remote monitoring capabilities. This makes it difficult for individuals and authorities to take immediate preventive actions when pollution levels exceed safe limits. Additionally, the absence of low-cost, portable, and user-friendly monitoring solutions restricts the ability to monitor indoor air quality effectively, which is equally important as outdoor air quality [6].

Contribution of the paper

This paper presents an IoT-based air pollution monitoring system with the following key contributions:

1. Design and development of a low-cost IoT-based air quality monitoring system capable of measuring temperature, humidity, and harmful gases including CO, NH₃, CH₄, and ethanol.
2. Integration of multiple sensors (DHT11, MQ5, MQ135) with a microcontroller for continuous real-time data acquisition and processing.
3. Implementation of a cloud-based monitoring platform for real-time visualization and remote access to environmental data using Wi-Fi connectivity.
4. Development of an alert mechanism using buzzer and LED indicators to notify users when pollutant levels exceed predefined safety thresholds.

Organization of the Paper

The remainder of this paper is organized as follows:

- **Section 2** presents the literature review, discussing existing IoT-based air quality monitoring systems and identifying their limitations.
- **Section 3** describes the system architecture and methodology, including the layered design, sensor integration, data processing, communication, and cloud-based monitoring framework.
- **Section 4** details the hardware implementation, including system components, circuit configuration, sensor calibration, and data transmission protocol.
- **Section 5** presents the experimental results and performance analysis, including system operation, data visualization using the cloud platform, alert mechanism validation, and comparison with expected air quality values.
- **Section 6** concludes the paper by summarizing the key findings, highlighting the effectiveness of the proposed system, and suggesting future enhancements such as machine learning integration and mobile application development.

AQI Category (Range)	PM ₁₀ 24-hr	PM _{2.5} 24-hr	NO ₂ 24-hr	O ₃ 8-hr	CO 8-hr (mg/m ³)	SO ₂ 24-hr	NH ₃ 24-hr	Pb 24-hr
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.6-1.0
Moderate (101-200)	101-250	61-90	81-180	101-168	2.1-10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	10.1-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748*	17.1-34	801-1600	1201-1800	3.1-3.5
Severe (401-500)	430+	250+	400+	748+*	34+	1600+	1800+	3.5+

Fig.1 AQI Category and amount of pollutants present in these range of AQIs

CENTRAL POLLUTION CONTROL BOARD'S AIR QUALITY STANDARDS	
AIR QUALITY INDEX (AQI)	CATEGORY
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-400	Very Poor
401-500	Severe

Fig. 2 Air Quality standards prescribed for the AQI ranges

Literature Review

Several research works have been conducted in the field of air quality monitoring using IoT technologies. These studies focus on different approaches including sensor-based monitoring, cloud integration, and smart city applications.

Jayakumar et al. [1] proposed an IoT-based monitoring system using Arduino and MQ-series gas sensors for real-time monitoring of air pollutants. While the system demonstrated feasibility and cost-effectiveness, it lacked predictive analysis capability. Gupta et al. [7] developed a smart city air monitoring system using an IoT network for large-scale environmental monitoring in urban areas, though the implementation had limited real-time alert functionality.

Singh et al. [8] implemented an Arduino-based gas detection system as a low-cost solution for detecting harmful gases. However, the system faced scalability limitations for large environments. Desai et al. [9] utilized BeagleBone Black for IoT monitoring, achieving high processing power and efficient data handling, but at increased cost and system complexity.

Phala et al. [10] proposed a standardized air quality monitoring system using standardized protocols for reliable monitoring, though without cloud-based visualization support. Pal et al. [11] developed an IoT-based pollution monitoring system for continuous environmental parameter monitoring but without AI-based analysis or intelligence.

Dhingra et al. [12] presented a mobile IoT-based monitoring system as a portable solution, though with limited hardware validation.

Anitha and Kumar [2] designed a smart air pollution prototype incorporating a PM sensor, gas sensors, and multiple filters with Raspberry Pi 3 B+ for both monitoring and air purification. The system effectively filtered particles above 0.3 μm and transmitted results to the ThingSpeak cloud platform. Kondeti et al. [4] presented a real-time air quality and weather monitoring system utilizing IoT for sustainable urban development, demonstrating the potential for comprehensive environmental surveillance.

From the literature review, it is evident that most existing systems focus on real-time monitoring using IoT sensors and microcontrollers. However, many lack efficient real-time alert mechanisms, cloud-based visualization, or scalability for widespread deployment. The proposed system addresses these gaps by combining low cost, real-time monitoring, cloud integration, and immediate alert mechanisms.

Problem Statement

Air pollution has become a significant environmental and health concern due to the presence of harmful gases such as carbon monoxide (CO), ammonia (NH₃), and methane (CH₄). Continuous exposure to these pollutants can lead to severe respiratory and cardiovascular diseases. However, existing air quality monitoring systems are often expensive, bulky, and limited to specific locations, making them inaccessible for widespread use.

Additionally, traditional systems lack real-time monitoring, remote accessibility, and immediate alert mechanisms, making it difficult for users to take timely preventive actions. Therefore, there is a need for a low-cost, portable, and IoT-based air pollution monitoring system that enables real-time data monitoring and provides alerts when pollution levels exceed safe limits.

System Architecture and Methodology

A. Proposed System Architecture

The proposed IoT-based air pollution monitoring system follows a layered architecture consisting of sensing, processing, communication, cloud, and application layers. The overall workflow is represented as:

Sensors → Microcontroller → Communication Module → Cloud Platform → Dashboard

B. Sensor Layer

The sensor layer is responsible for collecting environmental data. The system employs:

- **DHT11 sensor** for temperature and humidity measurement
- **MQ5 sensor** for detecting methane and combustible gases
- **MQ135 sensor** for detecting ammonia, carbon monoxide, ethanol, and other air pollutants

These sensors continuously monitor environmental conditions and generate electrical signals corresponding to pollutant levels. The MQ-series sensors operate on the principle of varying electrical resistance in response to target gas concentrations.

C. Processing Layer

The Arduino Uno microcontroller serves as the processing unit. Its functions include:

- Collecting raw data from sensors via analog and digital input pins
- Converting analog signals into digital values using the onboard ADC
- Comparing readings with predefined threshold values
- Preparing formatted data packets for transmission

D. Communication Layer

The ESP8266 Wi-Fi module establishes internet connectivity and transmits sensor data to the cloud platform. Data transmission utilizes the HTTP protocol, with the module sending periodic updates to the ThingSpeak API.

E. Cloud Layer

The ThingSpeak cloud platform provides:

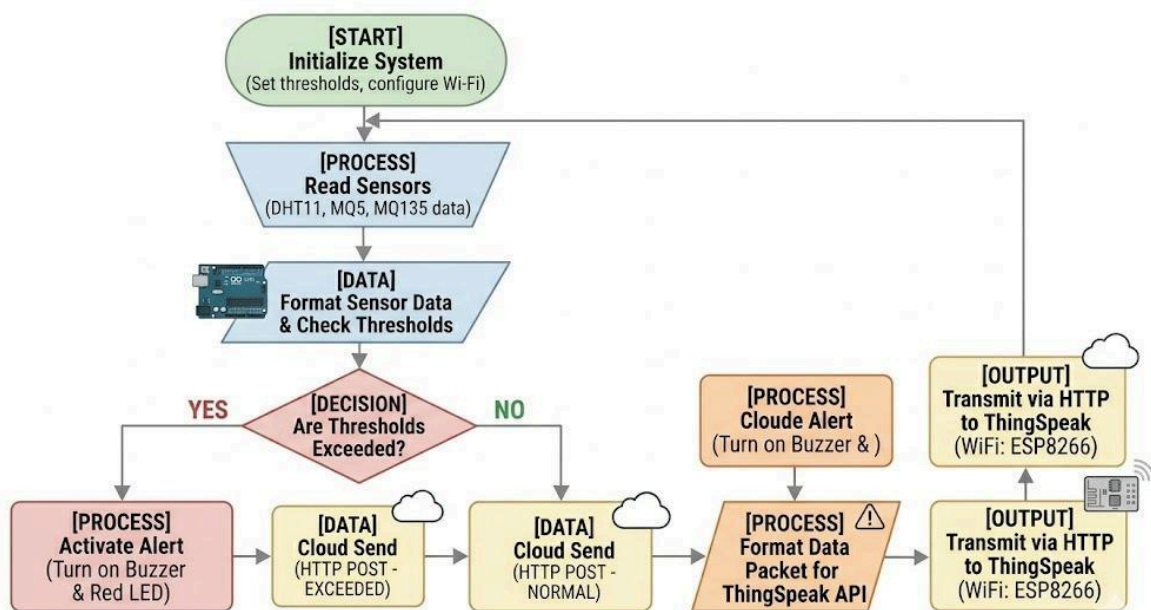
- Data storage and management
- Real-time visualization through graphical plots
- API access for data retrieval and analysis

Users can access environmental data remotely through a web interface from any location with internet connectivity.

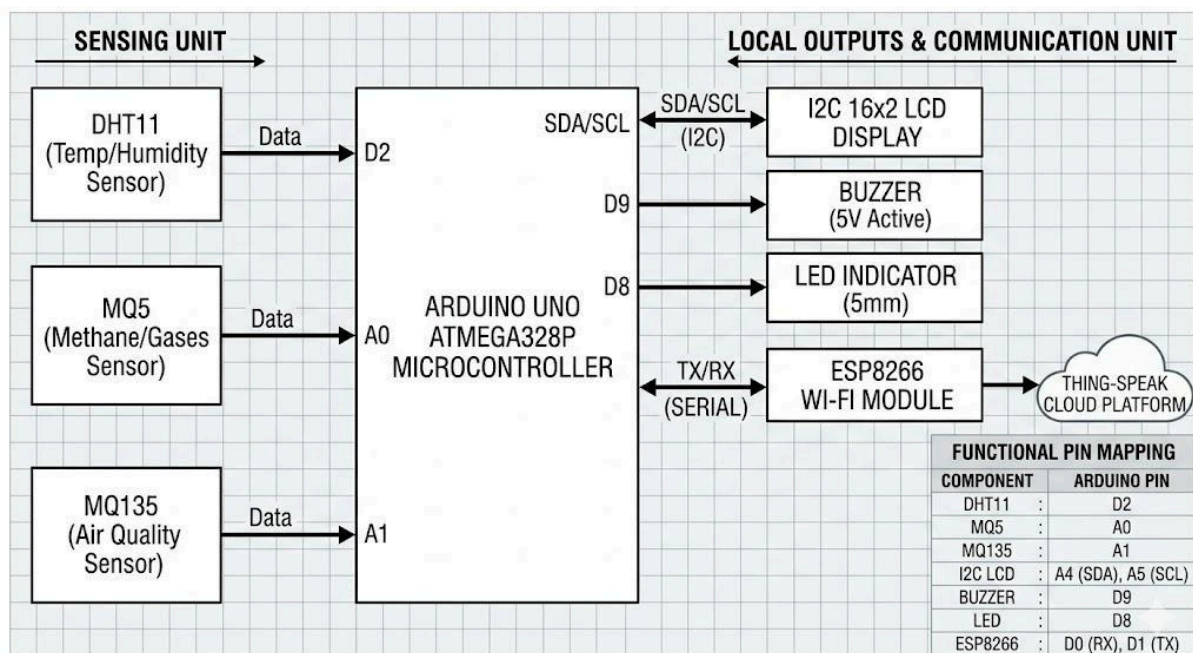
F. Application Layer

The application layer provides:

- Graphical representation of sensor data
- Real-time monitoring dashboard
- Alert system using buzzer and LED indicators when pollutant levels exceed safety thresholds



ALGORITHM FLOWCHART



HARDWARE SYSTEM ARCHITECTURE

Hardware Implementation

A. System Components

Table I summarizes the hardware components used in the system implementation.

TABLE I: Hardware Components

Component	Specification	Purpose
Arduino Uno	ATmega328P	Data processing and control
MQ5 Sensor	Analog output	Detect methane and combustible gases
MQ135 Sensor	Analog output	Detect CO, NH ₃ , ethanol, and air pollutants
DHT11 Sensor	Digital output	Measure temperature and humidity
ESP8266	Wi-Fi 802.11 b/g/n	Data transmission to cloud
LCD Display	16×2 characters	Local data visualization
Buzzer	5V active	Audible alert
LED	5mm	Visual alert indicator

B. Circuit Configuration

The sensors are connected to the Arduino through analog and digital pins. The MQ5 and MQ135 sensors connect to analog inputs A0 and A1 respectively, while the DHT11 sensor connects to a digital pin. The ESP8266 module interfaces via serial communication pins. The LCD

display connects through I2C or parallel interface for real-time local output.

C. Sensor Calibration

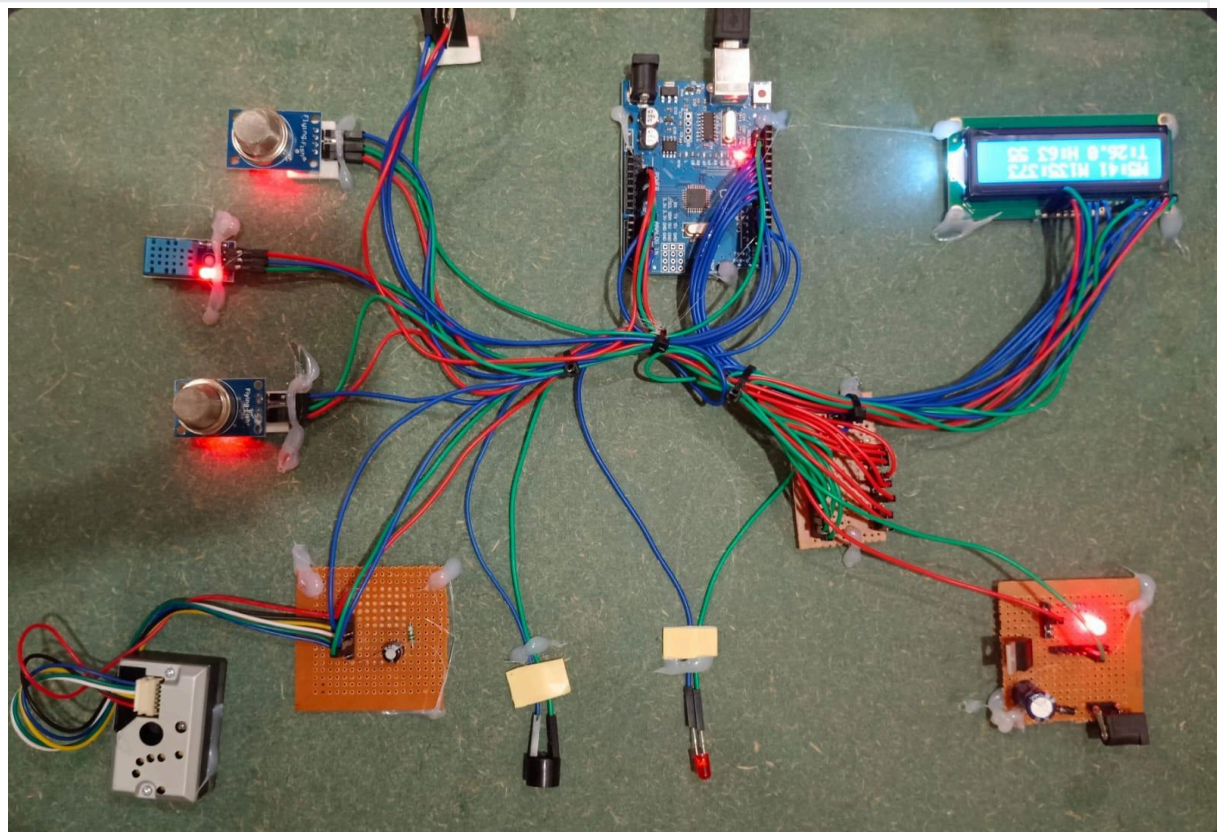
Proper calibration ensures accurate measurements:

- MQ sensors require a preheating period of 24 hours for initial calibration
- Calibration coefficients are determined based on manufacturer specifications
- Threshold values are set according to safe environmental limits defined by air quality standards

D. Data Transmission Protocol

Data transmission follows these specifications:

- HTTP protocol for cloud communication
- ThingSpeak API for data upload
- Update interval of 15 seconds to comply with ThingSpeak free tier limitations
- Data fields include temperature, humidity, and individual gas concentrations in parts per million (ppm)



CIRCUIT DIAGRAM

Results and Discussion

A. System Operation

The system was tested under real-time conditions to validate its functionality. Upon power-on, the system performs the following sequence:

1. Initializes sensor modules and establishes Wi-Fi connectivity
2. Connects to the local network via mobile hotspot
3. Authenticates with the ThingSpeak platform
4. Begins continuous data acquisition and transmission

B. Data Visualization

The ThingSpeak dashboard displays four graphical plots representing:

- Methane concentration (ppm) versus time
- Carbon monoxide and ethanol concentration (ppm) versus time
- Ammonia concentration (ppm) versus time
- Temperature (°C) and humidity (%) versus time

The system also supports geolocation tagging by specifying latitude and longitude coordinates, enabling location-specific data presentation.

C. Performance Analysis

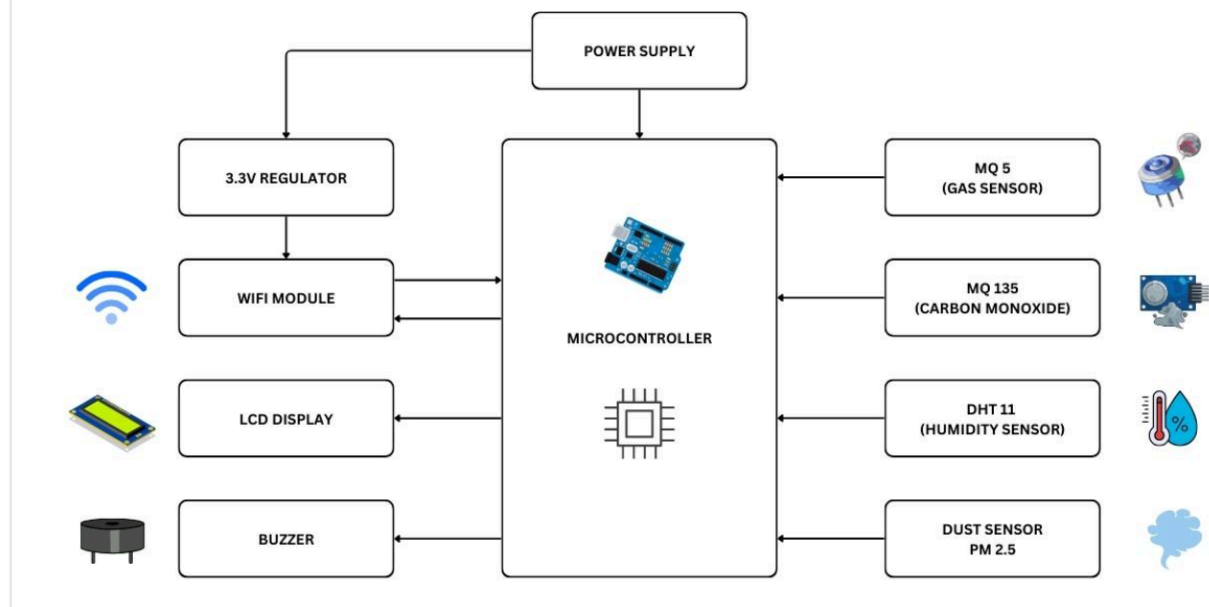
Experimental results demonstrate:

- Reliable sensor data acquisition with consistent readings
- Stable Wi-Fi connectivity and successful data transmission
- Real-time visualization with minimal latency
- Effective alert triggering when threshold values are exceeded

The graphical plots clearly show variations in environmental parameters over time, enabling users to identify trends and anomalies in air quality conditions.

D. Alert Mechanism Validation

The alert system was tested by exposing sensors to elevated gas concentrations. When pollutant levels exceeded predefined thresholds, the buzzer activated and LED indicators illuminated, providing immediate user notification.



BLOCK DIAGRAM

The results are shown in the pictures below:

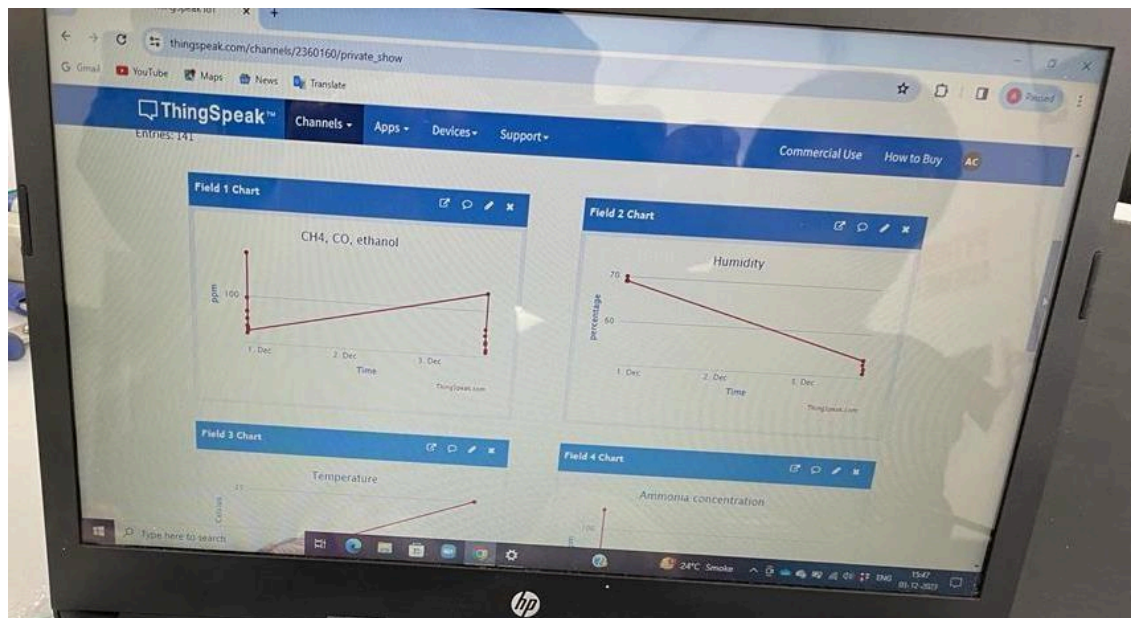


Fig.3: Graphical Plots of Methane, Carbon Monoxide & Ethanol (in ppm) and Humidity (in percentage) in surrounding air

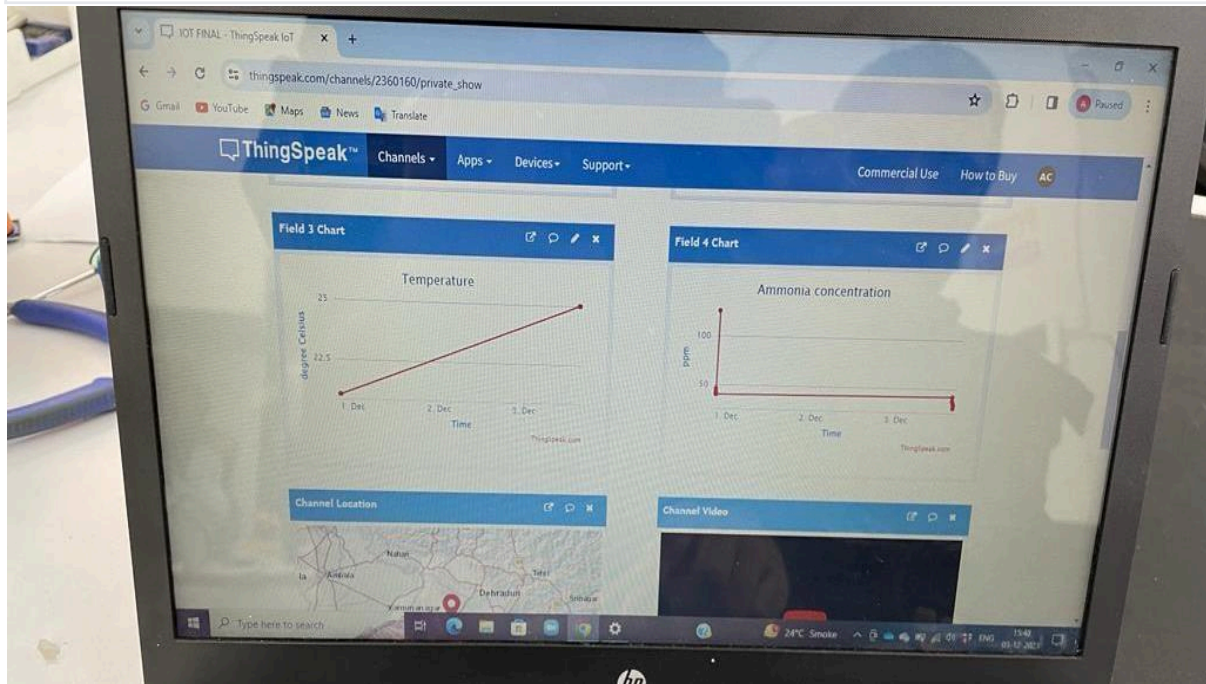


Fig.4: Graphical plots of Temperature (in degree Celsius) and Ammonia Concentration (in ppm) in surrounding air

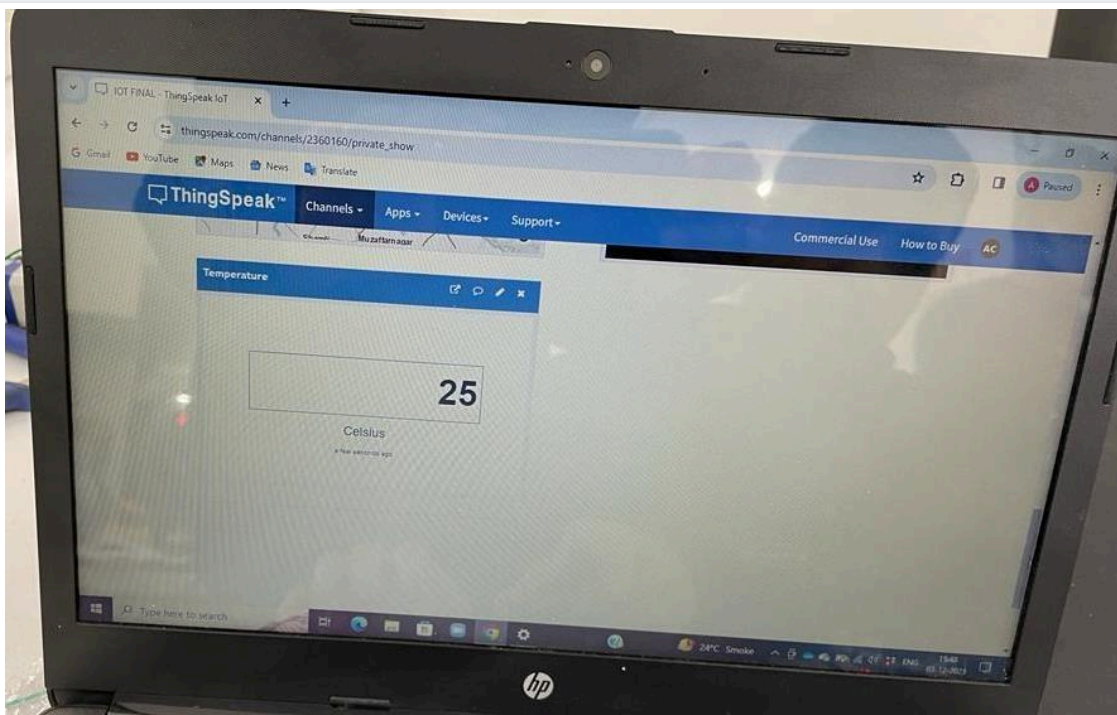
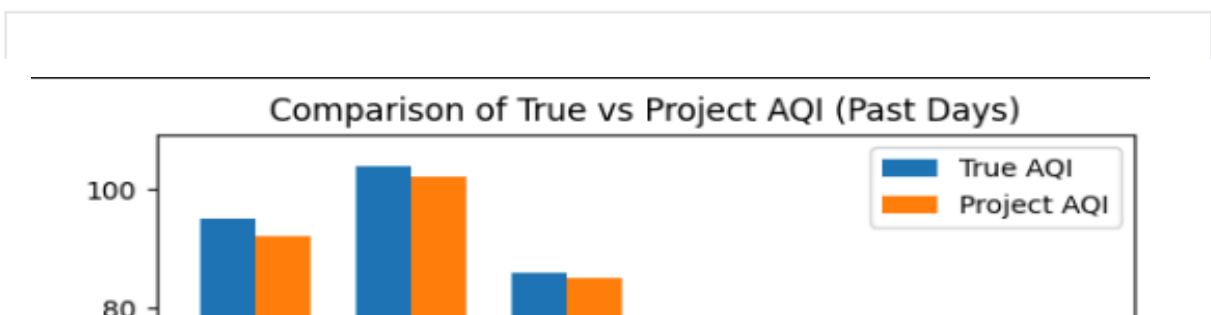


Fig.5: Data Showing the current temperature of the room in degree Celsius



COMPARISON TABLE

Parameter	Expected Values	True Values	Unit	Observations
AQI	60	62	Index	Very close
Temperature	35	34.8	°C	Accurate
Humidity	24	26	%	Slight variation
CO ₂	420 – 480	~450	ppm	From MQ135
CO	0.5 – 1.5	~1.0 (estimated)	ppm	Safe level
NH ₃	5 – 20	~10 (estimated)	ppb	Very low
LPG / Methane	1 – 5	~3 (estimated)	ppm	No leakage
MQ135	Not available	312	Analog	Air quality sensor
MQ5	Not available	148	Analog	Gas sensor



SENSOR DATA GRAPH

The bar chart illustrates a comparison between the **actual AQI values** obtained from standard air quality sources and the **AQI values calculated by the IoT-based air pollution monitoring system** over the past few days.

From the chart, it can be observed that the **project values closely follow the trend of the actual AQI values**. On days where the actual AQI increases or decreases, the project readings also show a similar pattern. This indicates that the system is effectively capturing changes in air quality.

Although there are **minor differences between the two sets of values**, these variations are expected due to factors such as **sensor calibration, environmental conditions, and the limitations of low-cost MQ sensors**. Unlike professional monitoring stations, MQ sensors provide approximate analog readings, which are then converted into AQI values.

Overall, the comparison demonstrates that the proposed system is **reliable and reasonably accurate for real-time air quality monitoring**, making it suitable for practical applications such as environmental awareness and IoT-based monitoring solutions.

Conclusion

This paper presents an IoT-based air pollution monitoring system designed to detect and analyze harmful gases in the surrounding environment. The proposed system effectively measures key parameters including temperature, humidity, and concentrations of toxic gases, providing real-time information about air quality. By continuously monitoring these parameters, the system identifies hazardous conditions and enables users to take necessary preventive actions.

The integration of IoT technology allows remote access to environmental data from any location through internet connectivity. Using Wi-Fi and mobile hotspot networks, the system ensures real-time data transmission and visualization on a cloud platform. The alert mechanism using buzzer and LED indicators enhances user awareness by providing immediate notifications when pollution levels exceed safe limits.

The proposed system is cost-effective, environmentally friendly, and easy to deploy, making it suitable for both indoor and outdoor applications. It contributes to improving public health by increasing awareness of air quality and reducing exposure to harmful pollutants. Furthermore, the system promotes the adoption of modern IoT technologies, encouraging further research and development in smart environmental monitoring.

Future work will focus on incorporating machine learning algorithms for predictive analysis, expanding the sensor array to include particulate matter detection, and developing a dedicated mobile application for enhanced user interaction.

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